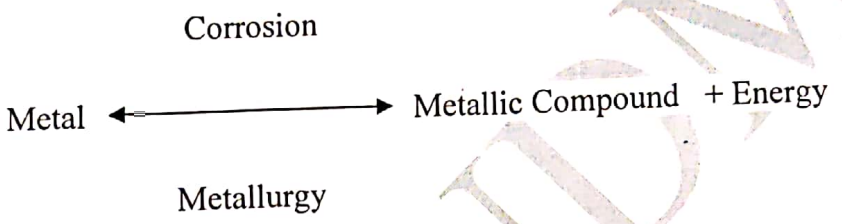
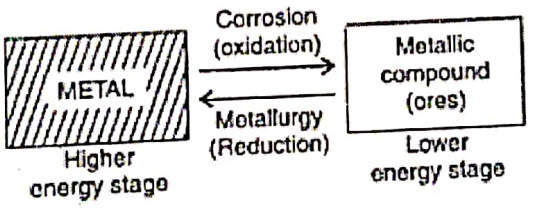
Part-II

Corrosion

Most of the metals are found in combined state in nature. The pure metal, in isolated state has higher energy and it is thermodynamically unstable, hence all the metals have tendency to go back to the combined state by the process, *Corrosion.*





Corrosion is slow, spontaneous destruction, or eating away of a metal due to a chemical or electro-chemical reaction between it and its environment. Eg. Rusting of Iron, formation of green layer over copper etc.

Theories of Corrosion

A. Dry or Chemical corrosion: The dry or chemical corrosion occurs through direct attack of gases like O_2, H_2, H_2S, Cl_2 etc.

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I. Corrosion by O_2

Direct attack of O_2 at low or high temperature causes corrosion. The extent of corrosion depends on the nature of oxide film formed over the surface of metal.

Mechanism:

i)Formation of metal ions

$$M \rightarrow M2+ + 2e- \quad \text{(Oxidation)}$$

ii) Oxygen changes to ionic form

$$\tfrac{1}{2} O_2 + 2e- \rightarrow O^{2-} \quad \text{(Reduction)}$$

iii) Metal – Oxide film formation

$$M + \tfrac{1}{2} O2 \rightarrow M^{2+} + O^{2-} = MO \quad \text{(Redox Reaction)}$$

Nature of the metal-oxide layer or film and corrosion

i) Stable oxide Layer

- Has fine-grained structure and gets tightly adsorbed to the metal surface.
- Impervious in nature and stops further O2 attacks through diffusion.
- ***Behaves as protective coating and no further corrosion can develop***

eg. Oxides of Al, Sn, Pb. Cu etc

ii) Unstable Oxide layer

- Produced on the surface of the noble metals.
- Decomposes back into metal and O_2.
- ***Oxidation corrosion is not possible***

eg. Ag, Au, Pt.

iii) Volatile Oxide layer

- Oxide layer volatilizes as soon as it is formed thereby leaving the underlying metal surface exposed for

further attack.

- ***Causes rapid, continuous and excessive corrosion.***

eg.MoO_3.

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iv) Porous Oxide Layer

• Layer has pores or cracks in the layer.

• The atm. O_2 has access to the underlying surface of the metal through the pores or cracks of the layer.

• ***Corrosion continues unobstructed till the entire metal is completely converted into its oxide.***

eg. Alkali and alkaline earth metals.

PILLING – BEDWORTH RULE

• **Definition: An oxide layer is protective or non-porous, if the volume of oxide is equal or more then the volume of the metal from which it is formed.**

On the other hand if the volume of oxide is less than the volume of the metal, the oxide layer is porous and non-protective, because it cannot prevent the access of O_2 to the fresh metal surface below.

II. <u>Corrosion by H_2</u>

(i)H_2 Embrittlement (Ordinary temperature)

$$Fe + H_2S \rightarrow FeS + 2H$$

$$H + H \rightarrow H_2$$

Formation of cracks and blisters on metal surface due to high pressure of H_2 gas

(ii)Decarburization (high temperature)

$$H_2 \rightarrow 2H$$

$$C + 4H \rightarrow CH_4$$

The process of decrease in carbon content in steel is termed as decarburization of Steel

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III. Liquid metal corrosion

Due to chemical action of flowing liquid metal on solid metal or alloy at high temp.

- It leads to either dissolution of solid metals by liquid metal or internal penetration of liquid metal into the solid metal thereby weakening of solid metal.
- Commonly occurs in devices used for nuclear power

IV. Corrosion by other gases: The gases like H_2S, Cl_2 etc. attack on metal surface and a layer of metallic chloride or sulphide is formed. Eg. Formation of protective layer of AgCl on silver, formation of volatile layer of $SnCl_4$ on tin and porous non-protective layer of FeSby attack of H_2S on Iron.

B. Electrochemical or Wet Corrosion: Wet corrosion occurs in two conditions

i) When two dissimilar metals are contact with each other and either immersed or dipped partially in a conducting solution. In two dissimilar metals **(Galvanic corrosion)**, the metal with high reduction potential in electrochemical series acts as a cathode while other acts as an anode. eg. In Fe-Cu couple corrosion of Fe occurs.

ii) When a metal is exposed to varying concentration of oxygen i.e. corrosion occurs due to existence of separate anodic and cathodic area or parts(**Electrochemical cell**) between which current flows through the conducting solution.

Mechanism of Electrochemical corrosion:

i) At Anodic part: Oxidation of metal occurs by liberation of electrons. The metal is destroyed either by dissolving or forming compounds (such as oxide, chloride etc)
Hence corrosion occurs at the anodic area.

$$M \rightarrow M^{n+} + ne^-$$

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ii)At Cathodic part: Reduction occurs with absorption of liberated electrons. The cathodic reaction depends on nature of the environment and occurs in two ways: (i) By evolution of H_2 (ii) By absorption of O_2

a) Acidic environment: Evolution of H_2 occurs.(fig 5.17a)

$$2H^+ + 2e^- \rightarrow H_2$$

- All metals above H_2 in electrochemical series undergo this type of corrosion.
- This reaction occurs when anodic area is very large whereas the cathode area is small.

Example: **Iron metal in contact with HCl**

At Anode:

$$Fe \rightarrow Fe^{2+} + 2e- \quad \text{(Oxidation)}$$

At Cathode:

$$2H+ + 2e- \rightarrow H_2 \quad \text{(Reduction)}$$

b) Neutral environment: Absorption of O_2 occurs and hydroxide ion formation takes place (fig5.17 b)

$$\frac{1}{2} O_2 + 2e- + H_2O \rightarrow 2OH^-$$

- Occurs when the base metals are in contact with neutral solution like water, NaCl or moisture with dissolved Oxygen.
- Anodic parts are very small compared to the cathode partsand leads to localized corrosion The best example is rusting of iron.

Example:

Rusting of Iron metal in contact with a neutral solution

At Anode:

$$Fe \rightarrow Fe^{2+} + 2e- \quad \text{(Oxidation)}$$

At Cathode: $\frac{1}{2} O_2 + H_2O + 2e- \rightarrow 2OH^-$ (Reduction)

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Net Corrosion reaction is:

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$$

If enough oxygen is present, ferrous hydroxide is easily oxidized to ferric hydroxide.

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_2$$

(yellow rust)

If the supply of oxygen is limited, the corrosion product may be even black anhydrous magnetite **(Fe_3O_4).**

The corrosion product, rust is deposited at or near the cathode part, because of rapid diffusion of small Fe^{2+} ion than the larger OH^- ions.

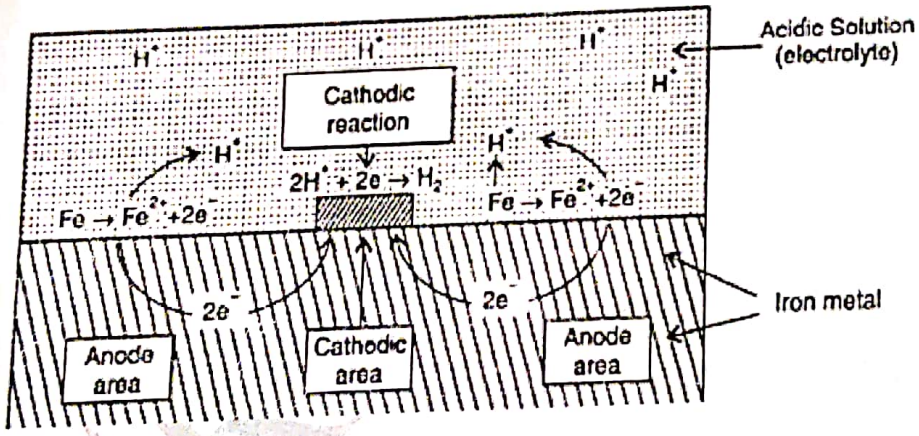


Fig 5.17a: wet corrosion with evolution of Hydrogen

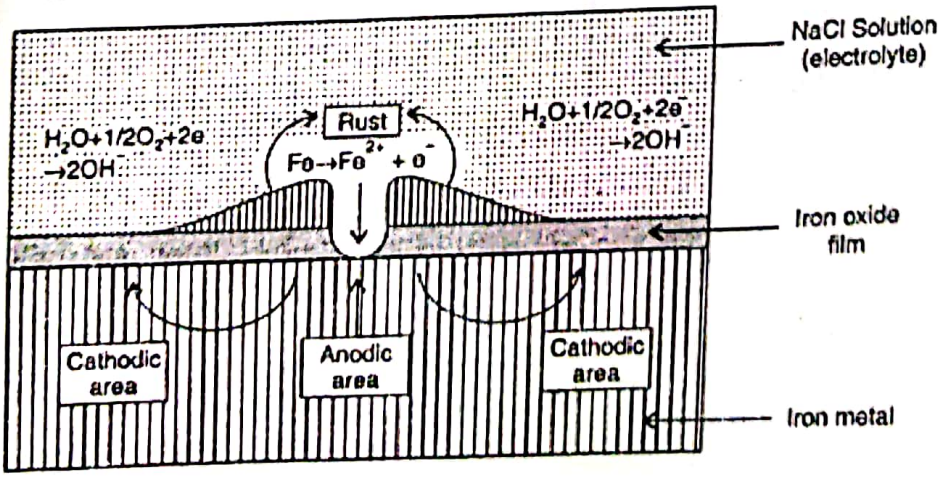


Fig 5.17b: wet corrosion with absorption of O_2

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Consequences of Corrosion: The process of corrosion is slow and occurs at the surface, but the losses incurred are enormous as destruction of machines, equipment results into following consequences..

- Plant shut down due to failure
- Requirement for replacement of corroded equipment
- Maintenance cost
- Loss of efficiency
- Contamination or loss of product (from the corroded container)
- Safety (fire due to release of toxic product or collapse of construction)
- Health (pollution)
- Necessity for over design to allow for corrosion
- ***Approximate estimate of loss due to corrosion is 2 to 2.5billion dollars per annum all-over the world***

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2. Corrosion by other gases -

→ gases like CO_2, SO_2, H_2S, F_2, Cl_2 etc. directly on metal surface.

→ Extent of corrosion depends on chemical affinity betn. metal and gas.
→ and nature of film formed
for eg. - AgCl forms Protective layer on Ag
- $SnCl_4$ is volatile, hence causes more Corrosion of Sn metal
- Steel + H_2S → FeS + H_2.
(At high temp.)

→ Corrosion by H_2 at ordinary temp and known as Hydrogen embrittlement.

→ due to the formn. of atomic hydrogen.

$Fe + H_2S \rightarrow FeS + 2H$.

As the vol. of atomic H increases it causes blisters and cracks. It also reacts with C to form Methane and make metal brittle.
(Decarburisation)

Liquid metal corrosion -

→ occurs in nuclear power plants where liquid and solid metals are in contact and solid metal dissolves in liquid metal or alloy at high temp

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Corrosion Control

Appropiate Design

- → no contact betn. two dissimilar metals
- → Diff. betn. E.P. is more
- → Anode should not be painted
- → Proper soff design
- → suitable Insulation
- → free circulation of air
- → no. Sharp corners or bends

To Prevent localised corrosion

Selection of metal

- → Pure metals
- → Homogenous alloys.
- → no tensile stress

Use of Corrosion Inhibitors

Cathodic Inhibitors

↓

Which slow down the rate of H^+ ions or Increase Hydrogen overVoltage or ~~Absorb~~ eliminating O_2

Organic Inhibitors = urea and thiourea or Arsenic or As_2O_3

Anodic Inhibitors Prevent the oxidation of anode → by forming insoluble films

eg. Chromate, Phosphate, molybdate, alkali and Tin organic Sod. Benzo-ate

by changing Env.

- → removal of Corrosive reagent
- → Deaeration
- → Neutralization of acid

P. coating

Protection of Electrical

Cathodic Protection

by

Anodic Protection

by forming protective layer of oxide.

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